FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2019

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 231**

General Instructions:

- The test consists of total 69 questions.
- Each subject (PCM) has 23 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-C & Section-D.**

  Section-A (01 – 03, 24 – 26, 47 – 49) *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 mark*** *for wrong answer.*

  Partial Marks **+1** for each correct option provided no incorrect options is selected.

  Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*

  Section-C (11 – 20, 34 – 43, 57 – 66) *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

  Section-D (21 – 23, 44 – 46, 67 – 69) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

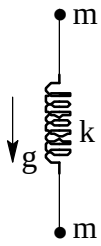
Physics

PART – I

SECTION – A
(Single Correct Choice Type)

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

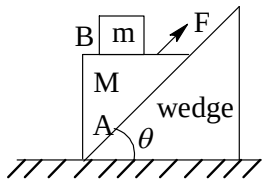
1. A mass m is hung on an ideal massless spring. Another equal mass is connected to the other end of the spring. The whole system is at rest. At t = 0, m is released and the system falls freely under gravity. Assume that natural length of the spring is L_0, its initial stretched length is L and the acceleration due to gravity is g. What is distance between masses as function of time?



(A) $L_0+(L-L_0)\cos\sqrt{\frac{2k}{m}}t$ (B) $L_0+(L-L_0)\cos\sqrt{\frac{k}{m}}t$

(C) $L_0-2(L+2L_0)\cos\sqrt{\frac{2k}{m}}t$ (D) $L_0+(L-L_0)\sin\sqrt{\frac{2k}{m}}t$

2. Wedge is fixed on horizontal surface. Triangular block A of mass M is pulled upward by applying a constant force F parallel to incline of the wedge as shown in the figure and there is no friction between the wedge and the block A, while coefficient of friction between A and block B of mass m is μ. If there is no relative motion between A and B then frictional force developed between A and B is

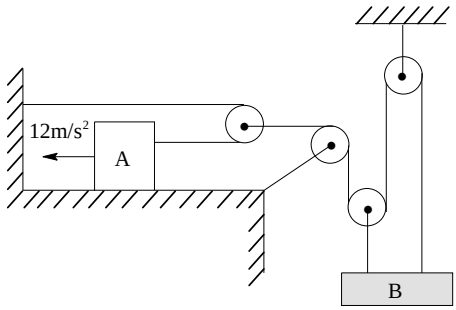


(A) $\left[\frac{F+(m+M)g\sin\theta}{(m+M)}\right]m\cos\theta$ (B) μmg

(C) $\left[\frac{F-(m+M)g\sin\theta}{(m+M)}\right]m\cos\theta$ (D) $\mu mg/2$

Space for Rough work

3. Assuming only translational motion of blocks A and B. Find the acceleration of B.



(A) 6m/s^2

(B) 2m/s^2

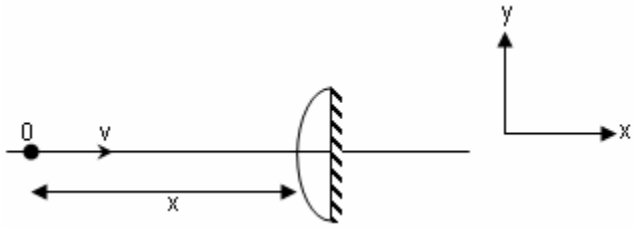
(C) 4m/s^2

(D) None of these

(One or More than one correct type)

This section contains **5 questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

4. When object 'O' moves towards a fixed lens mirror combination, select correct choice/choices:



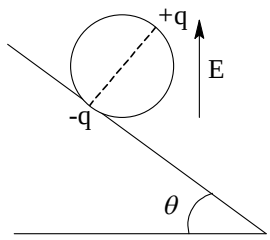
(A) Image moves towards negative x-axis

(B) Speed of image and object may be same

(C) Image may move faster than object

(D) Image may come closer to arrangement

5. Two point charges $+q$ and $-q$ are fixed on diametrically opposite point on the uniform ring, such that $-q$ is at bottom and is in contact with a perfect insulator incline plane. Total mass is equal to m. It remains in equilibrium on the rough inclined plane in the presence of uniform vertical electric field $\vec{E}$. Then



(A) The value of friction force is $mg\sin\theta$

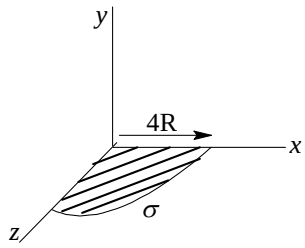
(B) The value of friction force is less than $mg\sin\theta$

(C) The value of electric field is $\dfrac{mg}{2q}$

(D) The value of electric field is $\dfrac{mg\tan\theta}{2q}$

Space for Rough work

6. Quarter non–conducting disc of radius 4R having uniform surface charge density σ is placed in xz-plane then which of the following is the correct:



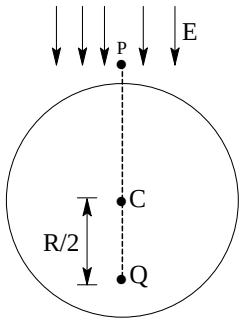
(A) Electric potential at (0, 3R, 0) is $\frac{\sigma R}{4\varepsilon_0}$

(B) Electric potential at (0,0,0) is $\frac{\sigma R}{2\varepsilon_0}$

(C) Electric field at (0, 3E, 0) is symmetric with x and z axis

(D) Electric field intensity at (-4R, 0, -4R) is equally inclined with x and z axis

7. A positive charge q is placed inside a neutral hollow conducting sphere of radius R, as shown in figure. Whole system is placed in uniform external vertical electric field pointing downward (line PCQ is also vertical) then select the correct statement(s) about electric field at point P. Point P is a point of the material inside the conductor.



(A) Electric field due to outer surface of sphere at point P is zero

(B) Magnitude of electric field due to inner surface of sphere at point P is $q/9\pi\varepsilon_0 R^2$

(C) Magnitude of electric field at point P, due to hollow sphere is less than E

(D) Direction of resultant electric field at point P due to charge q and hollow sphere is vertically upward

8. The particle displacement of a travelling longitudinal wave is represented by S = f (x, t). The midpoints of a compression zone and an adjacent rarefaction zone are represented by the letter 'C' and 'R'. Which of the following is true?

(A) $\left|\frac{\partial S}{\partial x}\right|_C = \left|\frac{\partial S}{\partial x}\right|_R$

(B) $\left|\frac{\partial S}{\partial t}\right|_C = \left|\frac{\partial S}{\partial t}\right|_R = 0$

(C) $(\text{pressure})_C - (\text{pressure})_R = 2\left|\frac{\partial S}{\partial x}\right|_C .(\text{bulk modulus of air})$

(D) Particles of air are stationary mid-way between 'C' and 'R'.

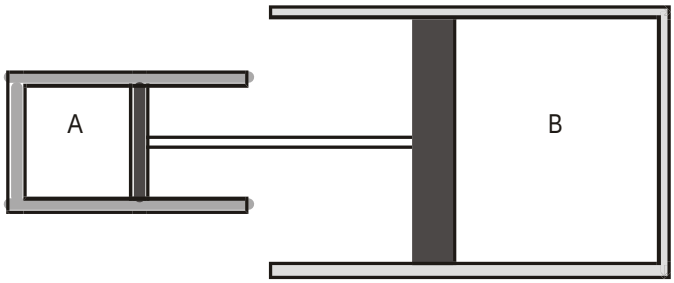
Space for Rough work

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Questions 9 & 10

Two fixed and horizontal cylinders A and B having pistons (both massless) of cross sectional area 100 cm^2 and 200 cm^2 respectively, are connected by massless rod. The piston can move freely without friction. The cylinder A contains 100 gms of an ideal gas (γ = 1.5) at pressure $10^5 N/m^2$ and temperature T_0. The cylinder B contains identical gas at same temperature T_0 but has different mass. The piston are held at the state such that volume of gas in cylinder A and cylinder B are same and is equal to $10^{-2} m^3$. The walls and piston of cylinder A are thermally insulated where as gas in cylinder B is maintained at constant temperature T_0. The whole system is in vacuum. Now the pistons are slowly released and they move towards left and mechanical equilibrium is reached at the state when the volume of gas in cylinder A becomes $25 \times 10^{-4}\ m^3$.



9. The mass of gas in cylinder B is
   (A) 200 gms (B) 600 gms
   (C) 500 gms (D) 1 kg

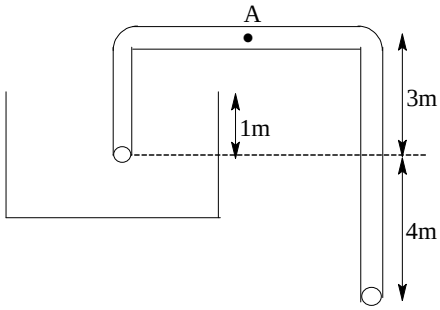
10. The change in internal energy of gas in cylinder A is :
   (A) 2000 J (B) 1000 J
   (C) 500 J (D) 3000 J

Space for Rough work

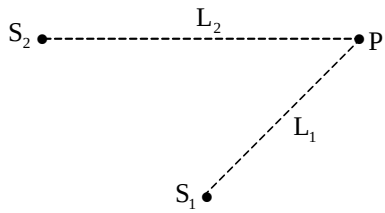
SECTION – C
(Single digit integer type)

This section contains **10 questions**. The answer to each question is a **Single Digit Integer** ranging from 0 to 9, both inclusive.

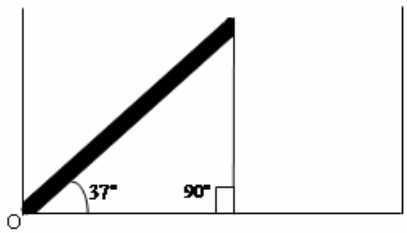
11. A liquid of specific gravity 0.5 is filled in a large container shown in figure. Area of cross section of tube is negligible as compare to the area of cross section of container then the magnitude of gauge pressure at point A is given by $\frac{x}{20}\text{atm}$ then x is :



12. Two sound sources shown in the figure vibrate in phase. By moving S_1 along PS_1 consecutive minima are heard when L_1-L_2 has values, 20cm, 60cm and 100 cm. If the frequency of sound source is $\frac{1700}{n}\text{Hz}$. Then find the value of n: [Speed of sound is 340 m/s]

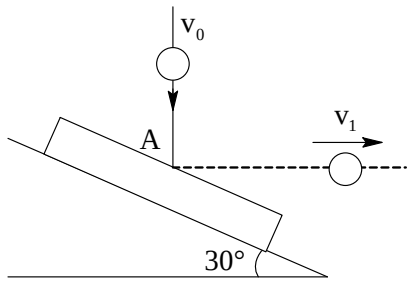


13. A cylindrical rod of uniform cross section, is attached at O in a water tank. The linear mass density of rod is $\lambda_0 x$, where x is distance of the element of the rod, from end O as shown in figure. If the tension in string is given by $\frac{10^4}{\text{P}}\text{N}$ then P is (Length of rod 1m, radius of area of cross section is $\frac{1}{\sqrt{\pi}}\text{m}, \rho_{\text{water}}=1000\text{kg m}^{-3}$, $g=10\text{ms}^{-2}, \lambda_0=10^3$ in S.I. unit)

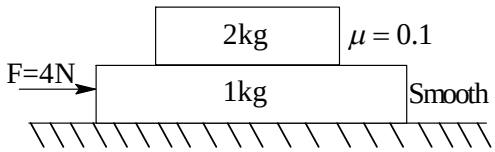


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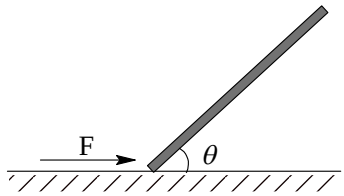
14. A steel ball falling vertically strikes a fixed rigid plate A with velocity v_0 and rebounds horizontally as shown. Assuming surface to be same and the effect of gravity on motion of ball to be neglected. If the coefficient of the restitution for the ball is 'e' then the value of $\frac{1}{e}$ is:



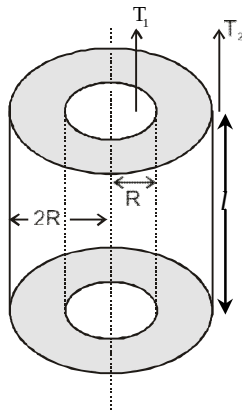
15. 2 kg block is kept on 1 kg block as shown. Both blocks are initially at rest. The friction between 1 kg block and fixed surface is absent and the coefficient of friction between 2kg block and 1 kg block is μ = 0.1. A constant horizontal force F=4 N is applied on 1 kg block. If the work done by the friction on 1 kg block in 2s is – XJ, then find the value of X.



16. A force F is applied to a uniform, thin rod of mass 4 kg and length $\ell = 50$ cm. The rod has pure translational motion in the vertical plane along a smooth, horizontal surface as shown. If F = 60 N, the angle θ for translation of rod in the given orientation comes to be $\theta = \tan^{-1}\left(\frac{2}{X}\right)$. Find X.



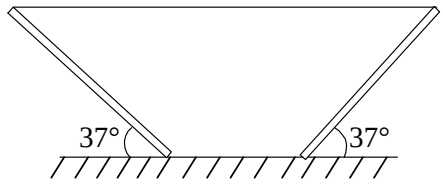
17. Inner surface of a cylindrical shell of length ℓ and of material of thermal conductivity k is kept at constant temperature T_1 and outer surface of the cylinder is kept at constant temperature T_2 such that $(T_1 > T_2)$ as shown in figure. Heat flows from inner surface to outer surface radially outward. Inner and outer radii of the shell are R and 2R respectively. Due to lack of space this cylinder has to be replaced by a smaller cylinder of length $\frac{\ell}{2}$ inner and outer radii $\frac{R}{4}$ and R respectively and thermal conductivity of material nk. If rate of radially outward heat flow remains same for same temperatures of inner and outer surface i.e. T_1 and T_2, then find the value of n.



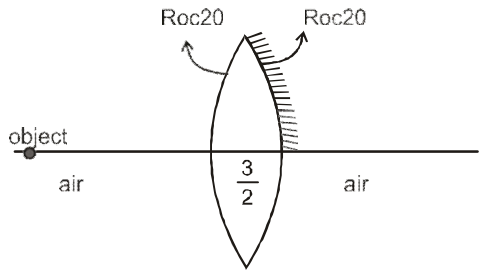
Space for Rough work

18. In a certain hypothetical radioactive decay process, species A decays into species B and species B decays
in to species C according to the given reactions.
A→2B + particles + energy
B→3C + particles + energy
The decay constant for species A is $\lambda_1 = 1 \text{ sec}^{-1}$ and that for species B is $\lambda_2 = 100 \text{ sec}^{-1}$. Initially 10^4 moles of species of A were present, while there were none of B and C. It was found that species B reaches its maximum number at a time t_0 = 2 ln (10) sec. Calculate the value of maximum number of moles of B.

19. Two uniform identical rods of same mass are tied together with the help of a string and balanced as shown in the figure. The minimum coefficient of friction for which the system will remain in equilibrium in the position is $\frac{2}{x}$. The value of 'x' is



20. Distance between two images formed by upper and lower part of the point object placed at 30 cm from given lens is (60 + x) cm, then the value of x is:

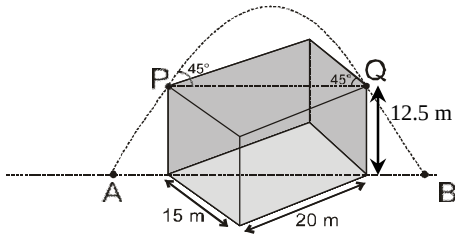


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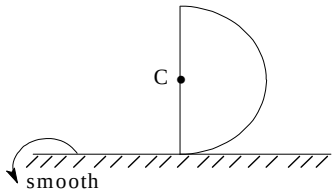
SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded ‑ off to the **second decimal place**; e.g. xxxxx.xx).

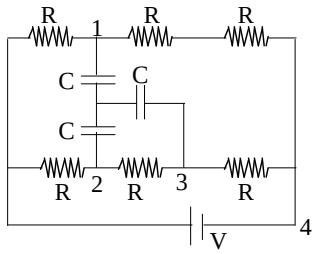
21. A commander fires a shell at certain angle of projection from 'A' which clears the building (cuboid) of dimensions 20 × 15 × 12.5 m^3 in its diagonal plane. Shell just clears the roof diagonally at points P and Q and falls on the other side of the building at B, and makes 45° angle with horizontal at P and Q as shown in the figure. If the range AB of the shell is $N\times\sqrt{3}$ metre then find N. (g = 10 m/s^2)



22. If angular acceleration of the uniform disc just after release from rest position shown in figure is $\frac{8\pi g}{NR}$, then calculate N. (C is centre of semi-circular disc.) ($\pi^2 = 10$)



23. In the show circuit, all three capacitor are identical and have capacitance $C\mu F$ each. Each resistor has resistance of $R\Omega$. An ideal cell of emf V volts is connected as shown. If the magnitude of potential difference across capacitor C_3 in steady state is $\frac{a}{b}V$ then value of $a\times b$ is:



Space for Rough work

Chemistry

PART – II

SECTION – A
(Single Correct Choice Type)

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

24. $\left[Ni\left(H_2O\right)_6\right]^{+2}$ complex typically absorbs at 600 nm. It is allowed to react with ammonia where a new complex $\left[Ni\left(NH_3\right)_6\right]^{+2}$ is formed, this complex should have absorption nearly at:

(A) 800 nm (B) 550 nm
(C) 650 nm (D) 600 nm

25. Which of the following statements is correct about behaviour of radicals?
(A) All radicals are very selective in their attack on other species.
(B) Radical reaction occur only in gas phase
(C) Radical may be produced by polar molecules like R – X in gas phase but not in aqueous phase.
(D) Radical can only be generated by non-polar molecules like alkanes

26. An amount of 1 mole each of A and D is introduced in 1L container, simultaneously the following two equilibria are established:

$A \rightleftharpoons B + C \quad K_C = 10^6 M^2$ & $B + D \rightleftharpoons A \quad K_C = 10^{-6} M^{-1}$

The equilibrium concentration of A will be:

(A) 10^{-6}M (B) 10^{-3}M
(C) 10^{-12}M (D) 10^{-4}M

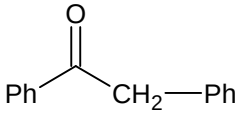
Space for Rough work

(One or More than one correct type)

This section contains **5 questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

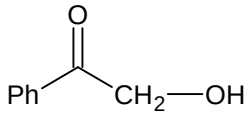
27. $N \equiv C\text{-}CH_2 - OCMe_3 \xrightarrow[(ii)\ H_2O,\ HI]{(i)\ PhMgBr}$

The major product(s) of above reaction ore:

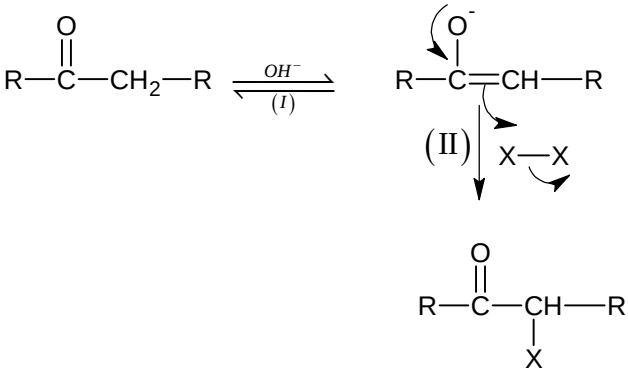
(A) 

(B) CH_3OH

(C) CMe_3I

(D) 

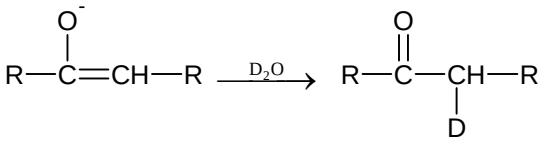
28. The correct statement(s) regarding the iodination of ketones is (are):



(A) Step (I) is slow and rate determining

(B) The reaction exhibit kinetic isotopic effect i.e. $k_H > k_D$

(C) Step (II) is slow and rate determining

(D) 

this occurs at similar rate as step (II)

Space for Rough work

29. Two moles of an ideal gas $(\gamma = 5/3)$ is subjected to following changes in state:

$$A(500k, 5bar) \xrightarrow[Isothermal\ expansion]{reversible} B \xrightarrow[cooling]{Isochoric} C(250k, 1bar) \xrightarrow[adiabatic\ compresion]{Single\ stage} D(3bar)$$

The correct statement(s) is (are):

(A) The pressure at B is 2.0 bar

(B) The temperature at D is 450 bar

(C) $\Delta H_{CD} = 1000R$ & $\Delta H_{AB} = \Delta U_{AB}$

(D) $\Delta U_{BC} = 375R \& \Delta H_{ABCD} \neq 0$

30. A concentrated solution of H_2SO_4 is added to saturated solution of sodium dichromate as:

$$Na_2Cr_2O_7 + H_2SO_4 \longrightarrow 'A' + Na_2SO_4 + H_2O$$

Which of the following statement(s) regarding 'A' is/are correct?

(A) 'A' is an orange solid and is an acidic oxide

(B) 'A' is a green solid and is an amphoteric oxide

(C) 'A' dissolves in alkalies forming a yellow coloured solution

(D) On heating 'A' loses oxygen & forms a green coloured compound.

31. Which of the following property(ies) are identical for body centered & face centered cubic crystal with same unit cell edge length.

(A) Distance between the nearest neighbours in BCC and distance between a tetrahedral void and octahedral void in FCC

(B) Coordination number of corner atom in FCC (if all tetrahedral voids are filled) and BCC

(C) Number of octahedral voids in FCC and number of tetrahedral voids in BCC

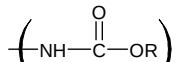
(D) Distance between second nearest neighbour in FCC & BCC

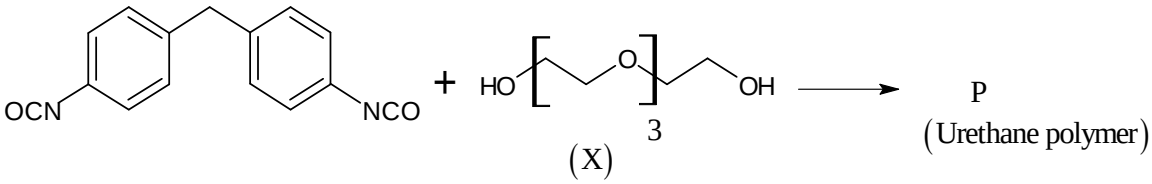
Space for Rough work

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

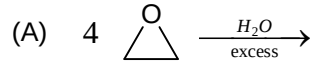
Paragraph for Questions 32 & 33

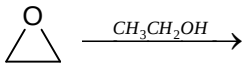
Isocyanates can undergo polymerisation with alcohols to form polymers containing  group and are called urethanes – hybrids between carbonates and urea i.e. half esters & half amides of carbonic acid. Consider the following polymerisation between diaryl diisocyanate with long chain aliphatic diol is:

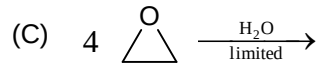


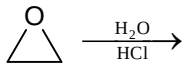
32. Which of the following is incorrect about the polymerisation reaction?
(A) It involves nucleophilic addition of alcohol on isocyanate.
(B) It involves nucleophilic addition elimination.
(C) On adding an amine RNH_2 to the polymer, the new functional group is urea rather than urethane.
(D) The amine added attack terminal isocyanate of the polymer.

33. The diol (X) can be synthesized by which of the reaction?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **10 questions**. The answer to each question is a **Single Digit Integer** ranging from 0 to 9, both inclusive.

34. The total energy of an electron in H-atom is given by

$E_n = \dfrac{-e^2}{8\pi \in_0 a_0 n^2}$ where n – shell number , a_0 – Bohr radius, $\in_0$ – Permittivity of the medium

An electron in a shell in H-atom has total energy $E = \dfrac{-e^2}{72\pi \in_0 a_0}$.

One of the subshell of this shell is described by radial part of wave function as

$\psi(r) = \dfrac{1}{81\sqrt{2}\pi}\left(\dfrac{1}{a_0}\right)^{3/2} \dfrac{r^2}{a_0^2} e^{-r/3a_0}$. The degeneracy of this subshell is:

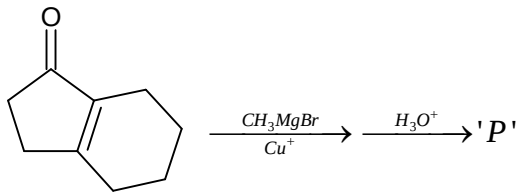
35. How many of the following species contain one or more lone pair(s) on central atom, that is present in an orbital having contribution from d-orbital.

SOF_4 $SeOCl_2$ SeF_3^+

XeF_5^+ BrF_6^- TeF_5^- ICl_4^-

36. The enthalpies of combustion of formaldehyde & paraformaldehyde (a cyclic polymer of formaldehyde) are -112 and -356 k cal/mol, respectively. The enthalpy of polymerisation of paraformaldehyde is $-92\,\text{k cal}$ per mole. If the molecular formula of paraformaldehyde is represented as $(CH_2O)_n$ then the value of n is:

Space for Rough work

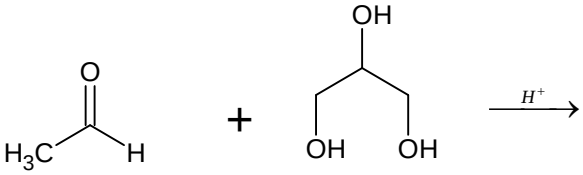
37. 

Number of $-OH$ groups in P is(are)?

38. Total number of molecules that can be hydrolysed at room temperature are:
XeF_4 NCl_3 NF_3 CCl_4 PCl_3 SF_6 TeF_6 $SeBr_4$

39. Dibromoacetic acid $(CHBr_2COOH)$ is oxidised to CO_2, H_2O & Br_2 by 12 equivalents of an oxidising agent. Same amount of acid can be neutralised by $'y'$ moles of NaOH. The value of 'y' is:

40. Number of optically active isomers formed by the given reaction is(are):



41. Number of sugars that give same osazone as given by D-glucose.
D-Mannose, D-Fructose, $D-$galactose, Starch, $D-$aldopentose

42. In a certain polluted atmosphere containing O_3 at a steady state concentration of 2×10^{-8}M, the hourly production of O_3 by all sources was estimated as 7.2×10^{-15}M. If only mechanism for the destruction of O_3 is of second order, the rate constant (in ml mol^{-1} s^{-1}) for the destruction reaction is:

43. Among the following, number of sols containing negatively charged colloidal particles is/are:
Soap solution of sodium stearate, sol of AgCl obtained by adding excess $AgNO_3$ to KCl solution, gold sol, $Fe(OH)_3$ sol, silcic acid sol, Basic dye, metal sulphide sol, acidic dye.

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. xxxxx.xx).

44. Fumaric acid $C_4H_4O_4$ present in 1g sample of mushroom was titrated using phenolphalein as indicator against OH^- ions which were obtained by electrolysis of water. As soon as OH^- ions are produced, they react with fumaric acid & at complete neutralisation immediately a pink colour is obtained. If electrolysis was done for 8685 seconds using 50mA current to reach end point, what was percentage of fumaric acid in mushroom.

45. The acid ionisation of hydrated alluminium ion is:

$$\left[Al\left(H_2O\right)_6\right]^{+3}\left(aq.\right)+H_2O\left(\ell\right)\rightleftharpoons\left[Al\left(H_2O\right)_5OH\right]^{+2}\left(aq.\right)+H_3O^+\left(aq.\right)$$

$K_a=1.0\times10^{-5}$

How many milligrams of $AlCl_3$ should be dissolved in sufficient water to get 400 ml of solution of pH, 3.0.

46. Two weak acids HA_1 & HA_2 each with same concentration & having pK_a values 3 & 5 are placed in contact with hydrogen electrode (1 atm 25°C) & are interconnected through a salt bridge. Emf of the cell in volt is: [Take $\frac{RT}{F}\left(2.303\right)=0.06$]

Space for Rough work

Mathematics

PART – III

SECTION – A
(Single Correct Choice Type)

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

47. If $P(\theta)$ and $Q\left(\frac{\pi}{2}+\theta\right)$ are two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,$ locus of the mid-point of PQ is:

(A) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{1}{2}$ (B) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=4$

(C) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=2$ (D) None of these

48. If A,B,C are acute positive angles, then $(\sin A+\sin B)(\sin A+\sin C)\frac{(\sin B+\sin C)}{\sin A\sin B\sin C}$ is:

(A) <8 (B) ≥ 8

(C) $=2$ (D) None of these

49. If $k>0, |z|=|\omega|=k,$ and $\alpha=\frac{z-\overline{\omega}}{k^2+z\overline{\omega}},$ then $\mathrm{Re}(\alpha)$ is equal to:

(A) $2k$ (B) $k/2$

(C) k (D) 0

Space for Rough work

(One or More than one correct type)

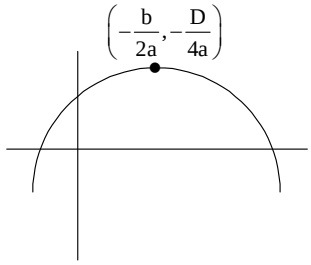
This section contains **5 questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

50. L_1 and L_2 are two lines whose vector equations are
$L_1 : \vec{r} = \lambda((\cos\theta + \sqrt{3})\hat{i} + (\sqrt{2}\sin\theta)\hat{j} + (\cos\theta - \sqrt{3})\hat{k})$
$L_2 : \vec{r} = \mu\left(a\hat{i} + b\hat{j} + c\hat{k}\right)$, where λ and μ are the scalars and α is the acute angle between L_1 and L_2. If the angle α is independent of θ, then the value of α is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

51. The graph of $f(x) = ax^2 + bx + c$ is



(A) $ac < 0$ (B) $bc > 0$

(C) $ab > 0$ (D) $abc < 0$

52. If P and Q are represented by the complex number z_1 and z_2 such that $|1/z_2 + 1/z_1| = |1/z_2 - 1/z_1|$ then

(A) ΔOPQ (Where O is the origin) is equilateral

(B) ΔOPQ is right angled

(C) The circumcenter of ΔOPQ is $\frac{1}{2}(z_1 + z_2)$

(D) The circumcenter of ΔOPQ is $\frac{1}{3}(z_1 + z_2)$

53. Let $(1+x^2)^2(1+x)^n = \sum_{k=0}^{n+4} a_k x^k$. If a_1, a_2 and a_3 are in arithmetic progression, then the possible value/values of n is/are

(A) 5 (B) 4

(C) 3 (D) 2

Space for Rough work

54. Consider the ellipse $\frac{x^2}{f\left(k^2+2k+5\right)}+\frac{y^2}{f\left(k+11\right)}=1$. If $f\left(x\right)$ is a positive decreasing function then

(A) The set of values of k for which the major axis is the $x-axis$ is (-3,2)

(B) The set of values of k for which the major $axis$ is the $y-axis$ is $\left(-\infty,2\right)$

(C) The set of values of k for which the major $axis$ is the $y-axis$ is $\left(-\infty,-3\right)\cup\left(2,\infty\right)$

(D) The set of values of k for which the major $axis$ is the $y-axis$ is $\left(-3,\infty\right)$

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Questions 55 & 56

A box has 10 coins. Five have heads on both sides. Three have tails on both sides; two are fair. A coin is chosen at random and tossed.

55. Probability that head appears is

(A) $\frac{4}{5}$ (B) $\frac{2}{5}$

(C) $\frac{3}{5}$ (D) $\frac{1}{5}$

56. Probability that the coin is fair given that head has appeared is

(A) $\frac{5}{6}$ (B) $\frac{1}{6}$

(C) $\frac{3}{8}$ (D) $\frac{5}{8}$

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **10 questions**. The answer to each question is a **Single Digit Integer** ranging from 0 to 9, both inclusive.

57. In equilateral triangle ABC of side p, a square is inscribed in its incircle. If area of the square is p^2 / k then value of *k* is

58. *f* is a function such that $f(x) + 2f(1-x) = x^2 + 1$. Value of $f(3)$ is

59. Degree of differential equation of family of ellipses of same size having minor axis along a given line is

60. Coefficient of x^6 in the expansion of $(1 + x + x^2 + x^3)^{-3}$ is

61. If the line $3x + 4y = 12$ intersects the hyperbola at *P* & P' and its asymptotes at *Q* and *Q* then value of $\frac{PQ}{P'Q'} + \frac{PQ'}{P'Q}$ is

62. Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, if let $(A + A^2 + A^3 + A^4 + \ldots + A^n) = 64$, then value of *n* is

63. If the number of the positive integral solution of equation $\left[\frac{x}{9}\right] = \left[\frac{x}{11}\right]$ is *n*, then $\frac{n}{3}$ is $\left([\cdot] \text{denotes GIF} \right)$.

64. $f(x) = \max \left| 2\sin y - x \right|$ Where $y \in R$ then minimum value of $f(x)$ is

65. If $f(n) = \frac{1}{\pi} \int_0^{\pi/2} \frac{\sin^2 n\theta}{\sin^2 \theta} d\theta$, $(n \in N)$, then the value of $\frac{f(15) + f(3)}{f(15) - f(9)}$ is

66. The sum to infinity of the series $\frac{5}{3^2.2} + \frac{7}{3^4.2} + \frac{1}{3^3.4} + \frac{11}{3^5.20} + \ldots..$ is K/9 then K is

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. xxxxx.xx).

67. The points of discontinuity of $y = g(u) = \dfrac{1}{u^2 + u - 2}$, where $u = \dfrac{1}{x-1}$ are x_1, x_2, x_3 find the value of $x_1 + x_2 + x_3 = ?$

68. If $\int_0^1 \dfrac{x^b - 1}{\ln x} dx = \ln(12.4)$, where $(b \geq 0)$ then value of 'b'

69. Three students appear in three examinations. The number of possible ways in which we can have result such that any students fails in all examinations nor neither these is a examination which is not cleared by any student is

Space for Rough work